Project-Definition Pack — Index

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

> **Project Identity**: A Kaggle-friendly, interpretable ML research pipeline for identifying candidate regulatory switch-like patterns in non-coding DNA, built on public datasets such as ENCODE.

This directory contains the complete project-definition pack. Each document is self-contained but cross-referential. Together they form the operating manual for the project.

Document Index

How to Use This Pack

1. **New contributors** — start with Document 12 (Onboarding Brief), then read Document 01 (Charter) and Document 11 (Glossary).
2. **Technical contributors** — additionally read Documents 03, 05, 06, and 08.
3. **Research contributors** — additionally read Documents 04, 05, and 06.
4. **Project leads / advisors** — read Documents 01, 02, 09, and 10 for governance and milestones.

Scope Lock (Reproduced for Visibility)

- **Domain**: Non-coding regulatory genomics.
- **Core Objective**: Discover and interpret candidate regulatory switch-like patterns in non-coding DNA.
- **Early Execution Environment**: Kaggle-friendly.
- **Early Model Philosophy**: Interpretable and lightweight first.
- **Large Genomic Transformer Training**: Out of scope for initial build because of compute limits.
- **Preferred Data Sources**: Public datasets such as ENCODE.
- **Long-Term Evolution**: Stronger models, richer interpretation, broader generalization — but only after a working baseline exists.

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